NIOSH RI 9672: Evidence for the Proximity Warning Proposal

The 2007 NIOSH report backs the proposal's main claims with field data: radar alone floods operators with false and nuisance alarms, tags miss untagged people, and NIOSH named sensor fusion, speed-adaptive range and lockout-tagout as the next steps. Notes below are grouped by the proposal item they support, with report page numbers for citing.

Source

Ruff, T. (2007). *Recommendations for Evaluating and Implementing Proximity Warning Systems on Surface Mining Equipment*. NIOSH Report of Investigations 9672, DHHS (NIOSH) Publication No. 2007-146. Spokane Research Laboratory, Spokane, WA.

The study ran from 2000 to 2006 with MSHA, Caterpillar and Phelps Dodge. Field tests were at the Phelps Dodge Morenci copper mine on Cat 793 and 797 haul trucks.

Item 6: Definition of the issue

Two-thirds of blind-area deaths happened behind reversing machines, and the tested radar gave a correct alarm only 40% of the time.

Accident evidence (Table 1, pp. 3–4). MSHA reports from January 2000 to mid-2006 list 21 US surface-mine deaths partly caused by blind areas.

- By machine: 38% dump trucks, 33% front-end loaders.
- By mine type: 29% coal, 38% stone, 29% sand and gravel.
- **67% happened in the rear blind area of a reversing machine.**

Breakdown of Table 1 (own count, not stated in the report):

Type of accident	All mines	Coal mines
Person on foot run over	11	1
Machine went over a dump edge, drop-off or highwall	8	4
Collision with a light vehicle	2	1
Total	21	6

The coal edge cases were 3 haul trucks backing over a dump point and 1 dozer over a highwall. This supports geofencing bench edges and dump yards. One light-vehicle collision killed 2 people.

Scale. Blind-area deaths were about 7% of all surface-mine deaths in 2000–2005 (p. 44). They happened even though audible back-up alarms were mandatory (p. 1).

Why industry hesitated (p. 1): too little field testing, high false-alarm rates, and poor reliability and heavy maintenance in mine conditions.

Gap (i): radar cannot tell objects apart (Table 2, p. 17). A Preco radar on a Cat 793B was logged 24 hours a day for 7 days while reversing.

- 618 reversing moves produced 580 alarms, so the radar alarmed on **94% of reversals**.
- Only **40% of alarms were true** (235); 18% were false (102) and 42% were nuisance alarms (243). None were missed.
- Drivers switched off the sound, and one display was found disconnected (pp. 21, 42).
- Passive sensors "will alarm often in the mining environment due to the detection of all large obstacles including rocks, berms, and deep ruts" (p. 47).

Need for radar or camera alongside tags (p. 47). Tag systems risk "increasing the potential of collisions with obstacles that are not outfitted with a tag. This is also true for GPS-based systems."

Gap (iii): fixed operation. "Ideally, detection range would automatically adjust with equipment speed, but this option was not available on the systems tested" (p. 38, repeated p. 47). A 7.6 m radar range "is only effective for slow moving situations" (p. 18).

Lockout-tagout and start-inhibit. "Most accidents happen after a person or vehicle approaches a stationary piece of equipment and the equipment moves unexpectedly" (p. 38).

Items 7, 8 and the R&D component

NIOSH's own future-research list (pp. 44–45) names sensor fusion, speed-adaptive detection and lockout-tagout, which are the core of Objectives 1 and 2.

Future research (pp. 44–45). NIOSH says two or more technologies need to be combined to:

1. provide redundancy;
2. give close-in detection at low speed and long-range detection at high speed;
3. let the operator check the alarm visually;
4. detect road edges;
5. cut nuisance and false alarms.

It names "the fusion of multiple sensors and visual information" as the next step. It adds that the same system "could also be used for security, tracking, hazard warning near stationary machines, lockout/tagout, and communications."

On combining sensors (p. 39). "An effective proximity warning system will require multiple technologies that combine obstacle detection and alarm functions with the ability to make a visual check of the blind area."

Tags (p. 38). Tags give two-way alarming, so both the driver and the tagged person are warned. They can also serve vehicle tracking and security check-in, which supports the unauthorised-entry feature.

Item 9: Benefits and cost

In 2007, tag systems cost 3–5 times as much as radar, so tags plus radar at radar-level cost would be a real contribution.

Comparison of technologies (Table 3, p. 37). Cost bands: Low = under $5,000 per machine, High = over $10,000.

Technology	Max range	Sensors for front + rear	Two-way alarm	False alarms	Nuisance alarms	Mud/dust tolerance	Install difficulty	Cost per machine
Sonar	3 m	4 or more	No	Medium	High	Low	Low	Low
Radar	7.6–17 m	2 to 4 or more	No	Medium	High	Medium	Low	Low to Medium
Magnetic-field tags	18 m	1 or 2	Yes	Low	Medium	High	Medium	Medium to High
Radio-frequency tags	80 m	2	Yes	Low	Medium	High	Medium	High

2007 prices (p. 44):

- Preco radar: about $3,200 for 4 sensors, cabling and a display. Adding cameras costs another $2,500–5,000.
- AMT tag system: about $15,600 per haul truck (2 RF units, 4 cameras, monitor, cabling), plus about $1,600 per light-vehicle tag.
- Field-test selection ceiling: under about $15,000 per truck (p. 11).

For comparison, the proposal's budget works out to about Rs 1.05 lakh per vehicle unit (Rs 8.40 lakh ÷ 8) and Rs 8,000 per tag (Rs 6.40 lakh ÷ 80).

Cost of a death (p. 44). A mine fatality costs a company about $2.5 million in 1990 dollars (Miller & Galbraith, 1995). This covers medical costs, lost wages, disruption, legal fees and lawsuit awards, but not fines.

Tag upkeep (p. 44). Tag systems "will require more attention to ensure all tags are operational and recharged." This supports the battery-life target.

Item 10.1: Methodology

The report gives a ready method for the Phase 1 blind-area study and many practical lessons for Phase 2 mounting, GNSS and cab alarms.

Phase 1: hazard and blind-area study

Method (pp. 4–5, Appendix A). A simplified ISO 5006 procedure, with two plots per machine. One shows the ground plane; the other shows a 1.5 m plane for a standing person. Areas visible in mirrors are marked separately.

What to monitor (pp. 5, 39):

Machine	Areas to monitor
Haul truck	Front, right side, rear. On the 280-tonne Euclid EH4500, the right-side blind area for a person runs past the plot's 24 m edge (Fig. A.3, p. 55).
Front-end loader	Rear, plus the front when the bucket is raised
Excavator / shovel	Side opposite the cab, rear, and the swing radius
Dozer (tracked)	Rear, front with the blade raised, and the sides, because it can spin quickly
Motor grader	Front and rear (sides fairly visible with mirrors)

Which machines to equip (pp. 38–39). Decide by risk analysis: blind-area size, past collisions by part of the machine, and light-vehicle and foot traffic nearby.

Sizing warning zones (p. 38). Use SAE J1741 with operator and pedestrian reaction times, top speed, braking distance and machine size. Short range suits a stationary machine about to move. Long range suits a light vehicle pulling out in front of a moving machine.

Zone sizes from the GPS trial (p. 28). Cat 797 haul truck: 30 m warning radius and 60 m detection radius. Dozer and service pickups: 20 m and 40 m. Icons changed from green to yellow to red as vehicles came closer.

Phase 2: building the system

Mounting (pp. 15, 18–19, 39–40):

- Mount sensors no higher than chest height. At 2.7 m, and even at 2.1 m, people could walk under the radar beam close to the truck.
- When the radar was tuned to ignore the rear tyres, it missed a person within 3 m of the rear axle.

- Angle the rear antennas inward about 15° so the beams cross behind the dual tyres.
- Use brackets with adjustable height and tilt.
- Don't weld on the axle housing; ask the manufacturer before welding or drilling anywhere.

Ruggedness (pp. 21, 40). Cables and connectors must handle heat, vibration, abrasion and high-pressure washing. Avoid mounting where tyres throw mud. Water got into one radar enclosure, which justifies the IP67 rating.

Radio near metal (pp. 22, 39). Detection changes near metal surfaces, so mounting takes trial and error. One tag system had unexplained uneven zones and a gap near a parked conveyor. This supports the on-site UWB and radar measurements.

Direction (p. 34). Magnetic tags give distance only, not direction, so several antennas are needed to tell front from rear. This supports four UWB nodes at the vehicle corners. Magnetic tags were still detected under the engine bay and wheel wells.

GPS/GNSS (pp. 27–30):

- Accuracy with differential GPS was 2–3 m observed (2–5 m expected); one test showed a 2 m error at 30 m.
- Positions were broadcast once a second over a 900 MHz IP radio.
- Signal reflections once shifted a vehicle's position by 15 m.
- Deep pit walls may block satellites; ground-based pseudolites were suggested.
- Fixed hazards (dump points, power lines, buildings) were stored as map points. This is a direct precedent for RTK/NavIC geofencing.

UWB radar (pp. 24–26). The first prototype, centred at 5.65 GHz, could not see close in (reliable only at 9.1–13.7 m). After changes it reached 15.2 m for a person and a pickup. A trial antenna for spotting drop-offs behind the truck looked promising.

Cameras (pp. 30–31). A stereo-camera prototype correctly flagged people and vehicles, targeting two or more updates a second. Challenges were hardening, cost and changing light. This supports the edge-AI camera.

Cab alarms and display (pp. 18, 40–43):

- Sound matters most, because drivers don't look at the display while reversing.
- Use graded alarms: 3 fast pulses once at the outer edge of the zone, continuous sound only within 3 m.
- Continuous pulsing annoyed drivers, who switched the sound off. A spoken "check video" prompt was suggested.
- Show the proximity alarm on the camera monitor so there is one display. Switch camera views automatically by gear.
- Usual haul-truck practice is three cameras: front, right side and rear.
- Place the display with operator input; avoid night glare and new blind spots, and isolate it from vibration.

No automatic braking (pp. 43, 48). NIOSH advises against braking from sensor input because of nuisance alarms. This matches the proposal's advisory design with start-inhibit only on stationary machines.

Objective 3 / Phase 3: a test protocol to reuse

Appendix B and the radar field trial give a complete test-course and shadow-mode protocol that the pilot can adopt almost as written.

Test-course settings (Appendix B, pp. 69–73):

Parameter	Value
Reliable detection zone	Target detected 100% of the time
Sporadic detection zone	Detected about 10–100% of the time; under 10% counts as outside
Alarm delay	Under 200 ms after relative movement starts
Person target	Real person, 172 ± 8 cm tall; optional second test at about 150 cm
Vehicle target	A light vehicle used on site, parallel to the machine's long axis
Test area	Flat and dry, no debris over 8 cm; 50 m clear in front of the sensor, 25 m clear each side
False-alarm run	Machine drives under 8 km/h for about 15 m through clear ground
Mapping grid	1 m spacing or less
Approach speed	Person about 5 km/h; vehicle under 8 km/h
Moving-machine check	Machine approaches a stationary target; note differences over 30 cm

Mapping steps (p. 71):

1. Start on the machine's centreline, outside the zone, and walk toward the sensor.
2. Mark where the alarm starts; back off and repeat until the point is consistent. Mark sporadic points in a different colour.
3. Keep walking and mark where the alarm stops.
4. Repeat on each grid line to the right, then to the left, until no detection occurs.
5. Record mounting position and settings once false alarms are minimised.

Alarm classes for field logging (pp. 11, 16–17):

Class	Definition	Example
True	An object over 30 cm high in the zone that needs action or extra caution	Person, vehicle, large rock; a dump-point berm counts as true
Nuisance	An object the operator already knows about that poses little risk	Backing up to a rock pile at the loading point
False	No object over 30 cm in the zone	Ruts, small berms, small rocks, foliage
Missed	An object in the zone but no alarm	—

Field logging (pp. 11, 16, 19). Record time-stamped video alongside the alarm log, 24 hours a day for 7 days. Log the time of each movement, the activity (loading, dumping, parking), the alarm class and a description. Digital video recorders worked; tape recorders lost data. This is the shadow-mode method.

Driver feedback (Appendix C, p. 76). An end-of-shift form asks about reliability, false-alarm frequency (frequent, occasional, very rare), missed alarms, collisions avoided, and camera use.

Acceptance criteria (pp. 10–11). NIOSH used these to pick systems for field trials:

- reliably detects a person and a light vehicle at 6 m or more, front and rear;
- gives an effective alarm with a low false-alarm rate;
- survives mine conditions;
- fits without major changes to the machine;
- costs under about $15,000 per truck.

Test on the actual machine (p. 47). Results from one machine type do not transfer to another; a system that suits haul trucks may not suit excavators.

WP5: SOP and training

A lack of training "can severely affect" operator acceptance (p. 48), so the SOP and training deliverable should cover these points (pp. 43, 48):

- Show workers and operators the detection zones for people and vehicles as diagrams, plus the camera views.
- Include blind-area diagrams in general safety training for all staff.
- Train operators on display brightness and alarm volume; operators who didn't know about these settings complained and resisted using the system.
- Set up a fault-reporting and maintenance process to avoid long downtime.
- Get operator feedback during installation so displays don't block views or other controls.
- State that the system adds to existing safe-operating procedures and does not replace them.

Where the paper cuts against the proposal

Three points need handling before reviewers raise them.

1. **Test dummy.** NIOSH found a real person gave the most accurate detection results, better than manikins, corner reflectors or metal spheres (p. 10; Ruff, 2002). The proposal's test rig uses a motorised dummy, so add real-person grid tests following Appendix B.
2. **Age of the source.** The report is from 2007, based on US conditions and 2007 dollars. Statements such as "speed-adaptive range not available" describe systems tested in 2000–2006, not today's market. Cite it as foundational work and pair it with recent sources.
3. **Accident reduction not proven.** NIOSH did not measure whether these systems reduce accidents and recommends a before-and-after study across several mines (p. 44). An 8-vehicle, 2-year pilot cannot show this either, so state plainly that the measures are detection rate, alert delay, false alarms and near-misses.